Mathematical Statistics

Practice problems for the SNU Statistics graduate entrance exam

Part I: 47 original problems (10 points each) Part II: 40 past exam problems (2020–2026, restated)

Solutions are in a separate booklet.

Notation: Exp(θ) has mean θ; Gamma(α, β) has shape α and scale β; $\chi^2_\alpha(r)$, $t_\alpha(r)$, $F_\alpha(r_1, r_2)$, z_α are upper-α points. A sequence Y_n is $O_p(1)$ if for every $\epsilon > 0$ there are M, N with $P(|Y_n| > M) < \epsilon$ for all $n \geq N$; $Y_n = o_p(1)$ means $Y_n \xrightarrow{p} 0$; $Y_n = O_p(a_n)$ means $Y_n/a_n = O_p(1)$.

Problem 1 **[10 pts]**

Analysis

(a) Using the mean value theorem, show that $\dfrac{x}{1+x} < \log(1+x) < x$ for all $x > 0$.

(b) Let $f_n(x) = nxe^{-nx}$ for $x \geq 0$. Find the pointwise limit f. Is the convergence uniform on $[0, \infty)$? On $[a, \infty)$ for a fixed $a > 0$?

(c) Show that $\int_0^1 f_n(x)\,dx \to \int_0^1 f(x)\,dx$, even though the convergence in (b) is not uniform on $[0, 1]$.

(d) Let $K \subset \mathbb{R}^k$ be compact and $g : K \to \mathbb{R}$ continuous. Prove that g is uniformly continuous on K.

Problem 2 **[10 pts]**

Joint, marginal and conditional distributions

(X, Y) is uniformly distributed on $D = \{(x, y) : 0 < x < 1,\ 0 < y < x^2\}$.

(a) Find the joint pdf and the marginal pdfs of X and Y.

(b) Find $E(Y \mid X = x)$ and $\mathrm{Var}(Y \mid X = x)$, and use them to compute EY and $\mathrm{Var}\, Y$.

(c) Compute $\mathrm{Cov}(X, Y)$. Are X and Y independent?

(d) Compute $P(Y > X/2)$.

Problem 3 **[10 pts]**

Transformations

Let $X, Y \overset{\text{iid}}{\sim} \text{Exp}(1)$.

(a) Find the joint pdf of $U = X/(X+Y)$ and $V = X+Y$. Show that they are independent and identify their distributions.

(b) Using (a), find $E(X \mid X+Y=v)$ and $E\big(X^2 \mid X+Y=v\big)$.

(c) Find the pdf of $W = X - Y$.

(d) Let $m = \min(X,Y)$ and $R = \max(X,Y) - \min(X,Y)$. Show that m and R are independent and find their distributions.

Problem 4 **[10 pts]**

Multivariate normal

Let $\mathbf{X} = (X_1, X_2, X_3)' \sim N_3(\mathbf{0}, \Sigma)$ with $\Sigma = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.

(a) Find the distribution of $X_1 - X_2 + X_3$.

(b) Find the constant c for which $X_2 - c(X_1 + X_3)$ and $X_1 + X_3$ are independent, and justify independence.

(c) Find the conditional distribution of X_2 given $(X_1, X_3) = (x_1, x_3)$.

(d) Find the mgf $M(t) = E\exp(tX_1X_3)$ and state where it is finite.

(e) What is the distribution of $\mathbf{X}'\Sigma^{-1}\mathbf{X}$? Justify.

Problem 5 **[10 pts]**

Stochastic order and limits

(a) Show that if $X_n \xrightarrow{d} X$, then $X_n = O_p(1)$.

(b) Show that if $X_n = O_p(1)$ and $Y_n = o_p(1)$, then $X_n Y_n = o_p(1)$.

(c) Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} U(0, \theta)$. Show that $n(\theta - X_{(n)}) \xrightarrow{d} \text{Exp}(\theta)$. Conclude that $X_{(n)} - \theta = O_p(n^{-1})$ and $\sqrt{n}(X_{(n)} - \theta) \xrightarrow{p} 0$.

(d) Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Without quoting the delta method, show $\sqrt{n}(\bar{X}_n^2 - \mu^2) \xrightarrow{d} N(0, 4\mu^2\sigma^2)$, using the decomposition $\bar{X}_n^2 - \mu^2 = 2\mu(\bar{X}_n - \mu) + (\bar{X}_n - \mu)^2$.

Problem 6 **[10 pts]**

MLE, MME and efficiency

Let $X_1, \dots, X_n$ be a sample from the Rayleigh density

$$f(x;\theta) = \frac{x}{\theta}\exp\Big(-\frac{x^2}{2\theta}\Big), \qquad x>0,\ \theta>0.$$

(a) Show that $X_1^2 \sim \mathrm{Exp}(2\theta)$. Find the MLE $\hat\theta$, and its mean and variance.

(b) Compute the Fisher information $I(\theta)$ and show that $\hat\theta$ attains the Cramér–Rao lower bound. Is $\hat\theta$ the UMVUE?

(c) Using $EX_1 = \sqrt{\pi\theta/2}$, find the method-of-moments estimator $\tilde\theta$ based on $\bar X$ and its limiting distribution.

(d) Compute $ARE(\tilde\theta, \hat\theta)$.

Problem 7 **[10 pts]**

Multinomial reparametrization

A sample of n individuals is classified into three genotypes with counts

$$(N_1, N_2, N_3) \sim \text{Multi}\big(n;\ \theta^2,\ 2\theta(1-\theta),\ (1-\theta)^2\big), \qquad 0 < \theta < 1.$$

(a) Find the MLE $\hat{\theta}$.

(b) Show that $2N_1 + N_2 \sim B(2n, \theta)$, and find $E\hat{\theta}$ and $\operatorname{Var}\hat{\theta}$.

(c) Find the Fisher information per individual and the limiting distribution of $\sqrt{n}(\hat{\theta} - \theta)$. Is $\hat{\theta}$ the UMVUE?

(d) With $n = 200$ and $(N_1, N_2, N_3) = (50, 90, 60)$, test at level 0.05 whether the three-cell model above fits, against a general trinomial, using Pearson's statistic. State the degrees of freedom.

Problem 8 **[10 pts]**

Sufficiency, completeness, UMVUE

Let $X_1, \ldots, X_n$ be a sample from $f(x;\theta) = e^{-(x-\theta)}$, $x > \theta$, $\theta \in \mathbb{R}$.

(a) Show that $X_{(1)}$ is sufficient, and find its distribution.

(b) Show that $X_{(1)}$ is complete.

(c) Find the UMVUEs of θ and of θ^2.

(d) Show that $X_{(1)}$ and $S = \sum_{i=1}^{n}(X_i - X_{(1)})$ are independent, and find the distribution of S.

Problem 9 **[10 pts]**

Most powerful and UMP tests

Let $X_1, \dots, X_n$ be a sample from the Pareto density $f(x;\theta) = \theta x^{-\theta-1}$, $x > 1$, $\theta > 0$, and let $T = \sum_{i=1}^{n} \log X_i$.

(a) Show that $2\theta T \sim \chi^2(2n)$.

(b) Find the most powerful level-α test of $H_0 : \theta = 1$ vs $H_1 : \theta = \theta_1$ for a fixed $\theta_1 > 1$.

(c) Show that the test in (b) is UMP for $H_0 : \theta \le 1$ vs $H_1 : \theta > 1$, and express its power function with a chi-square cdf.

(d) For $n = 10$ and $\alpha = 0.05$ ($\chi^2_{0.95}(20) = 10.851$), give the rejection region and the power at $\theta = 2$.

Problem 10 **[10 pts]**

Likelihood ratio test

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\theta_1)$ and $Y_1, \ldots, Y_m \overset{\text{iid}}{\sim} \text{Exp}(\theta_2)$ be independent samples. Consider $H_0 : \theta_1 = \theta_2$ vs $H_1 : \theta_1 \neq \theta_2$.

(a) Show that $2 \log \Lambda$, the LRT statistic, is a function of $B = n\bar{X}/(n\bar{X} + m\bar{Y})$, and describe the form of the rejection region.

(b) Find the exact null distribution of B, and of $\bar{X}/\bar{Y}$.

(c) State the limiting null distribution of $2 \log \Lambda$ as $n, m \to \infty$.

Problem 11 **[10 pts]**

Signed rank test

Let $X_1, \dots, X_n$ be a sample from a continuous distribution symmetric about θ, and $W^+ = \sum_i \mathbf{1}\{X_i > 0\} R_i$, where R_i is the rank of $|X_i|$ among $|X_1|, \dots, |X_n|$. Under $H_0 : \theta = 0$:

(a) Show that $W^+ \overset{d}{=} \sum_{j=1}^{n} j B_j$ with $B_j \overset{\text{iid}}{\sim} \text{Bernoulli}(1/2)$. (You may use that the signs and the ranks of $|X_i|$ are independent.)

(b) Find EW^+ and $\operatorname{Var} W^+$, and the exact null distribution of W^+ for $n = 4$.

(c) Show that $W^+ = \sum_{1 \le i \le j \le n} \mathbf{1}\{X_i + X_j > 0\}$.

(d) The ARE of the signed rank test relative to the t test is $12\sigma^2 \big(\int f^2 \big)^2$, where f is the density of $X_1 - \theta$ and σ^2 its variance. Evaluate it for f uniform on $(-1, 1)$ and for $f(x) = \frac{1}{2} e^{-|x|}$.

Problem 12 **[10 pts]**

Bayes and minimax

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$ with prior $\theta \sim \text{Exp}(\beta)$, i.e. $\text{Gamma}(1, \beta)$. Let $T = \sum X_i$ and $s = (n + 1/\beta)^{-1}$.

(a) Find the posterior distribution of θ.

(b) Find the Bayes estimator under squared error loss, and write it as a weighted average of $\bar{X}$ and the prior mean.

(c) Under the loss $L(\theta, a) = (a - \theta)^2/\theta$, show that the Bayes estimator is $a^* = sT$ and that its posterior expected loss is s for every T.

(d) Show that $\bar{X}$ is minimax for the loss in (c).

Problem 13 **[10 pts]**

Probability inequalities

(a) Using Jensen's inequality, show that $(E|X|^r)^{1/r}$ is nondecreasing in $r > 0$.

(b) Let $\hat{p}$ be the proportion of successes in n Bernoulli(p) trials. Using Chebyshev's inequality, find n that guarantees $P(|\hat{p} - p| \geq 0.05) \leq 0.05$ for every p.

(c) (Chernoff bound) For $X \sim \text{Poisson}(\lambda)$, show $P(X \geq 2\lambda) \leq \inf_{t>0} e^{-2\lambda t} E e^{tX}$, and evaluate the bound.

Problem 14 **[10 pts]**

Conditional expectation and random sums

Let $N \sim \text{Poisson}(\lambda)$ be independent of $X_1, X_2, \ldots$, which are i.i.d. with mean μ, variance σ^2 and mgf M_X. Let $S = \sum_{i=1}^{N} X_i$ ($S = 0$ if $N = 0$).

(a) Find ES and $\operatorname{Var} S$.

(b) Show that $M_S(t) = \exp\{\lambda(M_X(t) - 1)\}$.

(c) If $X_i \sim \text{Bernoulli}(p)$, identify the distribution of S.

(d) Show that $E(X_1 \mid S, N = n) = S/n$ for $n \geq 1$.

Problem 15 **[10 pts]**

Moment generating functions and cumulants

(a) Identify the distribution with mgf $(0.3 + 0.7e^t)^5$.

(b) $X_1 \sim \text{Poisson}(2)$ and $X_2 \sim \text{Poisson}(1)$ are independent and $Y = X_1 + 2X_2$. Find the mgf, mean and variance of Y. Is Y Poisson?

(c) Show that the r-th cumulant of $\text{Gamma}(\alpha, \beta)$ is $\alpha\beta^r(r-1)!$, and find its skewness.

(d) A random variable satisfies $EX^k = (k+1)!$ for $k = 0, 1, 2, \ldots$. Identify its distribution.

Problem 16 **[10 pts]**

Uniform order statistics

Let $U_{(1)} < \cdots < U_{(n)}$ be the order statistics of $n \geq 2$ i.i.d. $U(0,1)$ variables.

(a) Find the joint pdf of $(U_{(1)}, U_{(n)})$.

(b) Show that the range $R = U_{(n)} - U_{(1)}$ has the Beta$(n-1, 2)$ distribution, and find ER.

(c) Using (b), show that $\mathrm{Cov}(U_{(1)}, U_{(n)}) = \dfrac{1}{(n+1)^2(n+2)}$.

(d) Find $P(U_{(1)} < \frac{1}{2} < U_{(n)})$.

Problem 17 **[10 pts]**

Normal samples and prediction

Let $X_1, \ldots, X_{n+1} \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, and let $\bar{X}$ and S^2 be computed from the first n.

(a) Show that $\dfrac{X_{n+1} - \bar{X}}{S\sqrt{1 + 1/n}} \sim t(n-1)$, and give a 95% prediction interval for X_{n+1}.

(b) Show that $ES = \sigma\sqrt{\dfrac{2}{n-1}}\dfrac{\Gamma(n/2)}{\Gamma((n-1)/2)}$.

(c) Find c such that $c\sum_{i=1}^{n}(X_i - \bar{X})^2$ minimizes $E\big(c\sum(X_i - \bar{X})^2 - \sigma^2\big)^2$.

Problem 18 **[10 pts]**

Chi-square, t and F

(a) If $T \sim t(r)$, show $T^2 \sim F(1, r)$.

(b) If $X \sim F(r_1, r_2)$, show $\dfrac{r_1 X}{r_2 + r_1 X} \sim \text{Beta}(r_1/2, r_2/2)$.

(c) $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ and $Y_1, \ldots, Y_m \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ independently. Show $\bar{X}/\bar{Y} \sim F(2n, 2m)$.

(d) Show that $EX = r_2/(r_2 - 2)$ for $X \sim F(r_1, r_2)$, $r_2 > 2$.

Problem 19 **[10 pts]**

Bivariate normal

(X, Y) is bivariate normal with means 0, variances 1 and correlation $\rho \in (-1, 1)$.

(a) Show that $P(X > 0, Y > 0) = \dfrac{1}{4} + \dfrac{\arcsin \rho}{2\pi}$.

(b) Show that $E \max(X, Y) = \sqrt{(1 - \rho)/\pi}$.

(c) Find $E(Y^2 \mid X = x)$ and $\mathrm{Cov}(X^2, Y^2)$.

Problem 20 **[10 pts]**

Quadratic forms

Let $\mathbf{Z} \sim N_n(\mathbf{0}, I)$.

(a) For $n = 3$, show that $Q = \frac{1}{2}(Z_1 - Z_2)^2 + \frac{1}{6}(Z_1 + Z_2 - 2Z_3)^2 \sim \chi^2(2)$ and that $Q = \sum_{i=1}^{3}(Z_i - \bar{Z})^2$.

(b) For symmetric A, show $E(\mathbf{Z}'A\mathbf{Z}) = \operatorname{tr} A$ and $\operatorname{Var}(\mathbf{Z}'A\mathbf{Z}) = 2\operatorname{tr}(A^2)$.

(c) Let $J = \frac{1}{n}\mathbf{1}\mathbf{1}'$. Show that $\mathbf{Z}'J\mathbf{Z}$ and $\mathbf{Z}'(I - J)\mathbf{Z}$ are independent, and identify their distributions.

Problem 21 **[10 pts]**

Modes of convergence

(a) Let $P(X_n = n) = 1/n$, $P(X_n = 0) = 1 - 1/n$. Show $X_n \xrightarrow{p} 0$ but $EX_n \not\to 0$.

(b) Let $Z \sim N(0,1)$ and $X_n = (-1)^n Z$. Show $X_n \xrightarrow{d} Z$ but X_n does not converge in probability to Z.

(c) Show that if $X_n \xrightarrow{d} c$ for a constant c, then $X_n \xrightarrow{p} c$.

(d) Show that if $E(X_n - a)^2 \to 0$, then $X_n \xrightarrow{p} a$.

Problem 22 **[10 pts]**

Central limit theorem in practice

(a) Approximate $P\big(\sum_{i=1}^{48} U_i > 26\big)$ for $U_i \overset{\text{iid}}{\sim} U(0,1)$.

(b) How large must n be so that a 95% interval $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ has half-width at most 0.03 regardless of p?

(c) Approximate $P(X \geq 120)$ for $X \sim \text{Poisson}(100)$ with a continuity correction.

Problem 23 **[10 pts]**

Delta method and variance stabilization

(a) For a Poisson(θ) sample, find the limit law of $\sqrt{n}(\sqrt{\bar{X}} - \sqrt{\theta})$ and use it to build an asymptotic 95% interval for θ.

(b) For an Exp(θ) sample, find the limit law of $\sqrt{n}(\log \bar{X} - \log \theta)$.

(c) For $\hat{p} = X/n$, $X \sim B(n, p)$, find the limit law of $\sqrt{n}(\arcsin \sqrt{\hat{p}} - \arcsin \sqrt{p})$.

Problem 24 **[10 pts]**

Cauchy location

Let $X_1, \dots, X_n$ be a sample from the Cauchy density $\frac{1}{\pi}\{1 + (x - \theta)^2\}^{-1}$.

(a) Using the characteristic function $Ee^{itX} = e^{i\theta t - |t|}$, show that $\bar{X}$ has the same distribution as X_1. Is $\bar{X}$ consistent?

(b) Find the limit law of $\sqrt{n}(\hat{m}_n - \theta)$, where $\hat{m}_n$ is the sample median.

(c) Compute the Fisher information $I(\theta)$ and the ARE of the median relative to an efficient estimator.

Problem 25 **[10 pts]**

A non-regular scale model

Let $X_1, \ldots, X_n$ be a sample from $U[\theta, 2\theta]$, $\theta > 0$.

(a) Find the MLE of θ and a minimal sufficient statistic.

(b) Find an unbiased estimator of θ that is a multiple of the MLE.

(c) Show that $(X_{(1)}, X_{(n)})$ is not complete.

Problem 26 **[10 pts]**

Two-parameter exponential

Let $X_1, \ldots, X_n$ ($n \geq 2$) be a sample from $f(x; \mu, \sigma) = \frac{1}{\sigma} e^{-(x-\mu)/\sigma}$, $x \geq \mu$.

(a) Find the MLEs $\hat{\mu}, \hat{\sigma}$.

(b) Show that $n(\hat{\mu} - \mu)/\sigma \sim \text{Exp}(1)$ and $2n\hat{\sigma}/\sigma \sim \chi^2(2n-2)$, independently.

(c) Given that $(X_{(1)}, \sum X_i)$ is complete sufficient, find the UMVUEs of σ and μ.

Problem 27 **[10 pts]**

Cramér–Rao bound

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$ and $\eta = e^{-\theta} = P(X_1 = 0)$.

(a) Find the Cramér–Rao lower bound for unbiased estimators of η.

(b) The UMVUE of η is $(1 - 1/n)^T$, $T = \sum X_i$. Find its variance and show that it exceeds the bound for all $\theta > 0$.

(c) Find the limit law of $\sqrt{n}(e^{-\bar{X}} - e^{-\theta})$ and compare.

Problem 28 **[10 pts]**

Normal with equal mean and variance

Let $X_1, \dots, X_n$ be a sample from $N(\theta, \theta)$, $\theta > 0$.

(a) Show that the MLE is $\hat{\theta} = \frac{1}{2}\big(-1 + \sqrt{1 + 4m_2}\big)$, where $m_2 = \frac{1}{n}\sum X_i^2$.

(b) Compute $I(\theta)$ and the limit law of $\sqrt{n}(\hat{\theta} - \theta)$.

(c) Compare $\hat{\theta}$ with $\bar{X}$ and S^2 by ARE.

Problem 29 **[10 pts]**

Kullback–Leibler divergence

(a) Compute $KL(\theta_1, \theta_0) = E_{\theta_0} \log\{f(X;\theta_0)/f(X;\theta_1)\}$ for $N(\theta, 1)$ and for Poisson(θ).

(b) Using $\log t \le t - 1$, show $KL \ge 0$ with equality iff the densities coincide.

(c) Explain why this suggests that the MLE is consistent.

Problem 30 **[10 pts]**

Rao–Blackwellization

(a) For a Poisson(θ) sample, compute $E\big(\mathbf{1}\{X_1 = 0\} \mid T\big)$, $T = \sum X_i$, and conclude.

(b) For a Bernoulli(θ) sample ($n \geq 2$), find the UMVUE of θ^2 by conditioning $X_1 X_2$ on T.

(c) For $U(0, \theta)$, improve $2X_1$ by conditioning on $X_{(n)}$.

Problem 31 **[10 pts]**

Basu's theorem

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ with σ^2 known.

(a) Use Basu's theorem to show that $\bar{X}$ is independent of S^2 and of $X_{(n)} - X_{(1)}$.

(b) Show that $E\big(X_{(n)} \mid \bar{X}\big) = \bar{X} + c_n\sigma$ for a constant c_n not depending on μ.

(c) Find $E(X_1^2 \mid \bar{X})$.

Problem 32 **[10 pts]**

Tests for a normal variance

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$.

(a) With μ known, find the UMP level-α test of $H_0 : \sigma^2 \leq \sigma_0^2$ vs $H_1 : \sigma^2 > \sigma_0^2$, and its power function.

(b) With μ unknown, show that the LRT of $H_0 : \sigma^2 = \sigma_0^2$ vs $H_1 : \sigma^2 \neq \sigma_0^2$ is based on $v = \hat{\sigma}^2/\sigma_0^2$ through $n(v - 1 - \log v)$, and describe the rejection region.

Problem 33 **[10 pts]**

Power and sample size

$X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\mu, 4^2)$; test $H_0 : \mu \le 10$ vs $H_1 : \mu > 10$ at $\alpha = 0.05$.

(a) Give the UMP test and its power function.

(b) Find the smallest n with power at least 0.9 at $\mu = 12$ ($z_{0.10} = 1.282$).

(c) With that n, compute the power at $\mu = 11$.

(d) Show that no UMP test exists for $H_0 : \mu = 10$ vs $H_1 : \mu \neq 10$.

Problem 34 **[10 pts]**

LRT for a uniform scale

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} U(0, \theta)$; test $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.

(a) Find the LRT statistic and the size-α rejection region.

(b) Show that under H_0, $2 \log \Lambda \sim \chi^2(2)$ exactly. Why is this not $\chi^2(1)$?

Problem 35 **[10 pts]**

Wald, score and LRT

$X \sim B(100, p)$ is observed as $x = 60$; test $H_0 : p = 0.5$ vs $H_1 : p \neq 0.5$ at level 0.05.

(a) Compute the LRT, Wald and score statistics.

(b) State the common limiting null distribution and the decisions.

(c) Which of the three needs only H_0, not the MLE? Give the corresponding 95% confidence set for p (the score interval) as an inequality.

Problem 36 **[10 pts]**

Pivots and confidence intervals

(a) For an $\mathrm{Exp}(\theta)$ sample, give an exact $1-\alpha$ confidence interval for θ.

(b) For a $U(0,\theta)$ sample, show that among intervals $[X_{(n)}/b, X_{(n)}/a]$ with $0 < a < b \le 1$ and coverage $1-\alpha$, the shortest is $[X_{(n)}, X_{(n)}\alpha^{-1/n}]$.

(c) For $N(\mu,\sigma^2)$ with both unknown, give an exact interval for σ.

Problem 37 **[10 pts]**

Bayesian prediction and risk

(a) Bernoulli(θ) trials with prior Beta(α, β) give t successes in n. Find $P(X_{n+1} = 1 \mid \mathbf{x})$. What is it for a uniform prior and $t = n$?

(b) $X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$, σ^2 known, prior $N(\mu_0, \tau^2)$. Show that the posterior mean has Bayes risk $(n/\sigma^2 + 1/\tau^2)^{-1}$, smaller than the Bayes risk σ^2/n of $\bar{X}$.

(c) In (b), find the frequentist MSE of the posterior mean as a function of θ, and show that it exceeds σ^2/n when $|\theta - \mu_0|$ is large.

Problem 38 **[10 pts]**

Wilcoxon rank sum

Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent samples from continuous distributions, $N = m+n$, and $W = \sum_j R(Y_j)$, the sum of the ranks of the Y's in the pooled sample.

(a) Under H_0 (same distribution), show that every rank R has $ER = \frac{N+1}{2}$, $\operatorname{Var} R = \frac{N^2-1}{12}$, and that two distinct ranks have covariance $-\frac{N+1}{12}$.

(b) Deduce $EW = \frac{n(N+1)}{2}$ and $\operatorname{Var} W = \frac{mn(N+1)}{12}$.

(c) For $m = n = 3$, find $P(W \geq 14)$ under H_0.

Problem 39 **[10 pts]**

Analysis: limits and series

(a) Using Problem 1(a), prove $\lim_{n\to\infty}(1 + x/n)^n = e^x$ for $x > 0$.

(b) Show that $\sum_{k=0}^{\infty} x^k/k!$ converges uniformly on $[-a, a]$ for every $a > 0$.

(c) Justify differentiating this series term by term, and conclude that its sum $E(x)$ satisfies $E' = E$.

(d) Show that a function with $f'' \geq 0$ on an interval is convex, using the mean value theorem.

Problem 40 **[10 pts]**

Linear algebra

(a) Show that a symmetric idempotent matrix has eigenvalues 0 or 1 and that its rank equals its trace.

(b) For X of size $n \times p$ with rank p, show that $H = X(X'X)^{-1}X'$ and $I - H$ are symmetric idempotent with traces p and $n - p$, and that $(I - H)X = 0$.

(c) Prove $(I + \mathbf{a}\mathbf{b}')^{-1} = I - \dfrac{\mathbf{a}\mathbf{b}'}{1 + \mathbf{b}'\mathbf{a}}$ when $1 + \mathbf{b}'\mathbf{a} \neq 0$, and $\det(I + \mathbf{a}\mathbf{b}') = 1 + \mathbf{b}'\mathbf{a}$.

(d) Use (c) to invert $\Sigma = (1 - \rho)I_k + \rho\mathbf{1}\mathbf{1}'$, and find for which ρ it is positive definite.

Problem 41 **[10 pts]**

UMVUE for a uniform region

$(X_1, Y_1), \ldots, (X_n, Y_n)$ are i.i.d. uniform on the square $\{(x, y) : |x| + |y| < \theta\}$, $\theta > 0$. Let $W_i = |X_i| + |Y_i|$ and $M = \max_i W_i$.

(a) Find the distribution of W_1, and show that M is sufficient.

(b) Show that M is complete.

(c) Find the UMVUEs of θ and of the area $2\theta^2$.

Problem 42 **[10 pts]**

Two-parameter uniform

$X_1, \ldots, X_n$ ($n \geq 2$) are i.i.d. $U(\theta_1, \theta_2)$, $\theta_1 < \theta_2$. You may use that $(X_{(1)}, X_{(n)})$ is complete sufficient.

(a) Find $EX_{(1)}$ and $EX_{(n)}$.

(b) Find the UMVUEs of $\theta_2 - \theta_1$, of $(\theta_1 + \theta_2)/2$, and of θ_1.

Problem 43 **[10 pts]**

UMP test for a uniform location

$X_1, \dots, X_n$ are i.i.d. $U(\theta, \theta + 1)$. Test $H_0 : \theta = 0$ vs $H_1 : \theta > 0$ at level $\alpha \in (0, 1)$.

(a) For a fixed $\theta_1 \in (0, 1)$, compute the likelihood ratio $p_{\theta_1}(\mathbf{x})/p_0(\mathbf{x})$ on each part of the sample space.

(b) Show that the test rejecting when $X_{(n)} \geq 1$ or $X_{(1)} \geq c$, with $c = 1 - \alpha^{1/n}$, has size α and is UMP.

(c) Find its power function.

Problem 44 **[10 pts]**

Non-uniqueness of UMP tests

$X_1, \dots, X_n$ are i.i.d. $U(0, \theta)$; test $H_0 : \theta \le \theta_0$ vs $H_1 : \theta > \theta_0$ at level α. Show that both

$$\phi_1 = \mathbf{1}\{X_{(n)} > \theta_0(1-\alpha)^{1/n}\} \qquad \text{and} \qquad \phi_2 = \mathbf{1}\{X_{(n)} > \theta_0\} + \alpha\,\mathbf{1}\{X_{(n)} \le \theta_0\}$$

are UMP, and that they have the same power function.

Problem 45 **[10 pts]**

LRT on a restricted parameter space

$X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\mu, 1)$ with parameter space $\mu \geq 0$. Test $H_0 : \mu = 0$ vs $H_1 : \mu > 0$.

(a) Find the MLE and show $2 \log \Lambda = n(\bar{X}^+)^2$, with $a^+ = \max(a, 0)$.

(b) Show that under H_0, $2 \log \Lambda$ has the mixture law $\frac{1}{2}\delta_0 + \frac{1}{2}\chi^2(1)$. Find the size-$\alpha$ critical value ($\alpha < \frac{1}{2}$) and compare with the naive $\chi^2(1)$ calibration.

(c) Find the limit law of $\sqrt{n}(\hat{\mu} - \mu)$ for $\mu > 0$ and for $\mu = 0$.

Problem 46 **[10 pts]**

Extremes

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(1)$.

(a) Show that $X_{(n)} - \log n \xrightarrow{d} G$ with $P(G \leq x) = \exp(-e^{-x})$ (Gumbel).

(b) Find the exact distribution of $nX_{(1)}$.

(c) Show that $nX_{(1)}$ and $X_{(n)} - \log n$ are asymptotically independent.

Problem 47 **[10 pts]**

Censored data

Lifetimes $X_i \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ are censored at a fixed time $c > 0$: we observe $Z_i = \min(X_i, c)$ and $\delta_i = \mathbf{1}\{X_i \le c\}$, $i = 1, \dots, n$.

(a) Write the likelihood and show that $(D, T) = (\sum \delta_i, \sum Z_i)$ is sufficient.

(b) Find the MLE of θ (when $D > 0$).

(c) Show that $I_n(\theta) = nP_\theta/\theta^2$ with $P_\theta = 1 - e^{-c/\theta}$, and give the limit law of the MLE. How much information is lost by censoring?

Part II. Past exam problems (restated)

The following are the mathematical statistics problems of the SNU Statistics graduate entrance exams from October 2020 to April 2026, restated in our own words with the original mathematical content. Regression-only problems are omitted. Tags give the exam date and problem number; the original papers are on the department website.

Past problem 1

[2020-10 #1]

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(0,1)$ with order statistics $X_{(1)} < \cdots < X_{(n)}$, $X_{(0)} = -\infty$, $X_{(n+1)} = \infty$. Let

$$T_n = \sum_{i=k+1}^{n} i \int_{X_{(n-i)}}^{X_{(n-i+1)}} \frac{\phi(x)}{1-\Phi(x)} dx, \qquad k < n.$$

(a) Find the distribution of T_n. (b) For $n = 2k$ and $k \to \infty$, find a_k, b_k such that the mgf of $a_k(T_n - b_k)$ converges to that of $N(0,1)$, and prove it.

Past problem 2 *[2020-10 #2]*

$X_i \sim N(\mu_i, 1)$, $i = 1, 2$, independent, with parameter space $\Omega = \{\mu_1 \geq 0, \mu_2 \geq 0\}$. Derive the LRT of $H_0 : (\mu_1, \mu_2) = (0, 0)$ vs $H_1 : (\mu_1, \mu_2) \neq (0, 0)$ and the condition determining its rejection region.

Past problem 3 *[2020-10 #3]*

For a sample from $N(\mu, \sigma^2)$ with σ^2 known, find the UMVUE of $e^{c\mu}$ ($c \neq 0$ known), and decide whether it attains the Cramér–Rao bound.

Past problem 4 *[2021-04 #3]*

$X_1, \ldots, X_n$ have pdf $\theta^{-1}e^{-x/\theta}$, $x > 0$, where the parameter space is $\theta \geq 1$. Find the MLE and the limit law of $\sqrt{n}(\hat{\theta} - \theta)$.

Past problem 5 *[2021-04 #4]*

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} U[0, \theta]$. Find a UMP test of $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.

Past problem 6 *[2021-04 #5]*

(X_i, Y_i), $i \leq n$, are i.i.d. uniform on the triangle $\{x > 0, y > 0, x + y < \theta\}$. Find the UMVUE of θ and its pdf.

Past problem 7

[2021-10 #3]

$X_1, \ldots, X_n$ have pdf $\sigma^{-1} e^{-(x-\mu)/\sigma}$, $x \geq \mu$, with both parameters unknown. Find the LRT of $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$.

Past problem 8 *[2021-10 #4]*

(X_i, Y_i) are i.i.d. uniform in the disk of radius r centred at the origin, and $\theta = r^2$. Find the UMVUE of θ and its pdf.

Past problem 9

[2021-10 #5]

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(0,1)$. Prove or disprove: $n\Phi(\min X_i)$ and $n\{1 - \Phi(\max X_i)\}$ are asymptotically independent.

Past problem 10

[2022-04 #1]

$X_1, \ldots, X_m \overset{\text{iid}}{\sim} U(0, \theta)$ and, independently, $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} U(0, \eta)$. Find the UMVUE of θ/η.

Past problem 11 *[2022-04 #2]*

A single X has pdf $e^{x-\theta}/(1+e^{x-\theta})^2$. Show that the UMP level-α test of $H_0 : \theta \leq 0$ vs $H_1 : \theta > 0$ rejects when $X > C$, and find C.

Past problem 12 *[2022-04 #3]*

Buses arrive at a stop as a Poisson process of rate 1 per hour, and each stays 10 minutes. Given that two buses arrived between noon and 1 pm, find the probability that both were at the stop at the same time.

Past problem 13

[2022-10 #1]

$X_1, \ldots, X_n$ have the logistic pdf $e^x/(1+e^x)^2$. Find the pdf of $\log \dfrac{1+e^{-X_{(1)}}}{1+e^{-X_{(n)}}}$.

Past problem 14

[2022-10 #2]

$\mathrm{Corr}(X_n, Y_n) \to 1$, and $(X_n - EX_n)/\sqrt{\mathrm{Var}\, X_n} \xrightarrow{d} X$. Show that $(Y_n - EY_n)/\sqrt{\mathrm{Var}\, Y_n} \xrightarrow{d} X$.

Past problem 15 *[2022-10 #4]*

$X_1, \ldots, X_n$ have the Pareto pdf $\theta a^\theta x^{-(\theta+1)}$, $x > a$, with $\theta > 0$ known. Find a level-α UMP test of $H_0 : a = a_0$ vs $H_1 : a \neq a_0$.

Past problem 16 *[2022-10 #5]*

$\mathbf{y} \sim N_n(\mu, V)$ with V positive definite, and A, B symmetric idempotent. Show that (1) $\mathbf{y}'A\mathbf{y}$ and $B\mathbf{y}$ are independent iff $BVA = 0$; (2) $\mathbf{y}'A\mathbf{y}$ and $\mathbf{y}'B\mathbf{y}$ are independent iff $AVB = 0$.

Past problem 17 *[2022-10 #6]*

$\mathbf{y} \sim N_n(\mathbf{0}, I)$ and X is $n \times k$ of rank $k < n$, $H = X(X'X)^{-1}X'$. Find the distributions of $\mathbf{y}'H\mathbf{y}$, $\mathbf{y}'(I-H)\mathbf{y}$ and $T = \mathbf{y}'H\mathbf{y}/\mathbf{y}'(I-H)\mathbf{y}$.

Past problem 18 *[2023-04 #1]*

For a random vector $\mathbf{y}$ with mean μ and covariance Σ and symmetric A, show $E(\mathbf{y}'A\mathbf{y}) = \operatorname{tr}(A\Sigma) + \mu' A \mu$.

Past problem 19

[2023-04 #3]

$U_1, U_2, \ldots \overset{\text{iid}}{\sim} U(0,1)$, independent of X with $P(X = x) = \frac{c}{x!}$, $x = 1, 2, \ldots$, $c = 1/(e-1)$. For $Z = \min\{U_1, \ldots, U_X\}$, compute $P(Z \le z)$, $0 < z < 1$.

Past problem 20

[2023-04 #4]

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$, $n > 1$. Show that $u(X_1) = \mathbf{1}\{X_1 \leq c\}$ is unbiased for $\Phi(c - \theta)$, and use it to find the UMVUE of $\Phi(c - \theta)$.

Past problem 21

[2023-04 #5]

$X_1, \ldots, X_n$ have the Weibull pdf $\lambda c x^{c-1} e^{-\lambda x^c}$, $x > 0$, c known. (1) Show that X_i^c is exponential. (2) Find the UMP level-α test of $H_0 : \lambda \geq \lambda_0$ vs $H_1 : \lambda < \lambda_0$.

Past problem 22

[2023-10 #1]

For a Poisson(λ) sample, estimate $\theta = e^{-\lambda}$ by $\hat{\theta}^{(1)} = e^{-\bar{X}}$ or $\hat{\theta}^{(2)} = \frac{1}{n}\sum \mathbf{1}\{X_i = 0\}$. Find both limit laws and decide which is more efficient.

Past problem 23 *[2023-10 #2]*

ϕ is the UMP size-α test of $H_0 : f = f_0$ vs $H_1 : f = f_1$, with power $\beta \in (0, 1)$. Find the UMP size-$(1 - \beta)$ test of $H_0^* : f = f_1$ vs $H_1^* : f = f_0$ in terms of ϕ.

Past problem 24 *[2023-10 #3]*

$X_1, \ldots, X_n$ are i.i.d. uniform on $(\theta, \theta + |\theta|)$, $\theta \neq 0$. Find the MME and the MLE of θ, and decide whether the MLE is consistent.

Past problem 25 *[2024-05 #1]*

For a normal sample, use $(n-1)S_n^2/\sigma^2 \sim \chi^2(n-1)$ to show $\sqrt{n-1}(S_n - \sigma) \xrightarrow{d} N(0, \sigma^2/2)$.

Past problem 26

[2024-05 #2]

X_i are i.i.d. with $P(X_i = \sigma) = P(X_i = -\sigma) = \frac{1}{2}$. Show that $S_n - \sigma = o_p\big((n-1)^{-\gamma}\big)$ for every $\gamma < 1$.

Past problem 27 *[2024-05 #3]*

For a sample from $U(\theta, \theta+1)$, $n \geq 2$, find a minimal sufficient statistic.

Past problem 28 *[2024-05 #4]*

$\mathbf{y} \sim N(X\beta, \sigma^2 I_n)$, with X of size $n \times (p+1)$ and full column rank, and H the hat matrix. Find the distribution of $F = \mathbf{y}'H\mathbf{y} / \mathbf{y}'(I-H)\mathbf{y}$.

Past problem 29 *[2024-10 #1]*

For $U(0,\theta)$ and $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ $(\theta_0 < \theta_1)$, show that any test with $\phi = 1$ when $x_{(n)} > \theta_0$ and $E_{\theta_0}\phi = \alpha$ is most powerful.

Past problem 30 *[2024-10 #2]*

If $X_n \xrightarrow{d} Z$, where all cdfs are continuous, show $X_n = O_p(1)$.

Past problem 31 *[2024-10 #3]*

Show that a compact subset of a metric space is complete.

Past problem 32

[2025-04 #1]

Let $X_n, Y_n = O_p(1)$ and $Z_n = o_p(1)$. Prove (1) $X_n + Y_n = O_p(1)$; (2) $X_n Y_n = O_p(1)$; (3) $Z_n X_n = o_p(1)$; (4) $Z_n = O_p(1)$.

Past problem 33

[2025-04 #2]

For a Beta$(\alpha, 1)$ sample: (1) show $\hat{\alpha}^{\mathrm{MME}} = \bar{X}/(1-\bar{X})$ and find its limit law; (2) find the MLE and its limit law; (3) compute the ARE of the MME with respect to the MLE.

Past problem 34

[2025-04 #3]

For the signed rank statistic $W_n = \sum \operatorname{sgn}(X_i - \theta_0) R(|X_i - \theta_0|)$ under $H_0 : \theta = \theta_0$ (symmetric continuous f): (1) show $W_n \overset{d}{=} \sum_j j S_j$ with i.i.d. fair signs S_j; (2) show $W_n / \sqrt{\operatorname{Var} W_n} \xrightarrow{d} N(0,1)$; (3) show $W_n^+ \overset{d}{=} \sum_j j B_j$ with $B_j \overset{\text{iid}}{\sim} \text{Bernoulli}(\frac{1}{2})$.

Past problem 35 *[2025-10 #1]*

$X_1, \ldots, X_n$ have pdf $2x/\theta^2$ on $(0, \theta)$. (a) Find the MLE and the MME. (b) Find their means and variances. (c) Compare them by MSE.

Past problem 36

[2025-10 #2]

$(X_1, X_2, X_3)' \sim N_3(\mathbf{0}, \Sigma)$ with $\Sigma = \begin{pmatrix} 1 & \rho & \rho^2 \\ \rho & 1 & \rho \\ \rho^2 & \rho & 1 \end{pmatrix}$, $|\rho| < 1$. (a) Find the joint mgf. (b) Find $\mathrm{Var}(X_1 - 2X_2 + X_3)$. (c) Are $X_1 - X_2$ and $X_2 - X_3$ independent?

Past problem 37 *[2025-10 #3]*

(a) State and prove the mean value theorem. (b) Give a function satisfying its hypotheses and explain. (c) Give a continuous function for which the conclusion fails, and explain why.

Past problem 38

[2026-04 #1]

$(X_1, X_2, X_3) \sim \text{Multi}(n; p_1, p_2, p_3)$, reparametrized by $\theta_1 = p_1$, $\theta_2 = p_2/(p_2 + p_3)$. (a) Find the MLE. (b) Find the Fisher information matrix. (c) Find the limit law of $(\hat{\theta}_1, \hat{\theta}_2)$.

Past problem 39

[2026-04 #2]

(X_1, X_2, X_3) is uniform on the unit ball. (a) Find the joint pdf of (X_1, X_2). (b) Find the marginal pdf of X_1. (c) Find the pdf of X_1 given $X_2 = 0$. (d) Find $\mathrm{Cov}(\mathbf{X})$.

Past problem 40

[2026-04 #3]

$f_n(x) = nx(1-x)^n$ on $[0,1]$. (a) Find the pointwise limit. (b) Is the convergence uniform?